Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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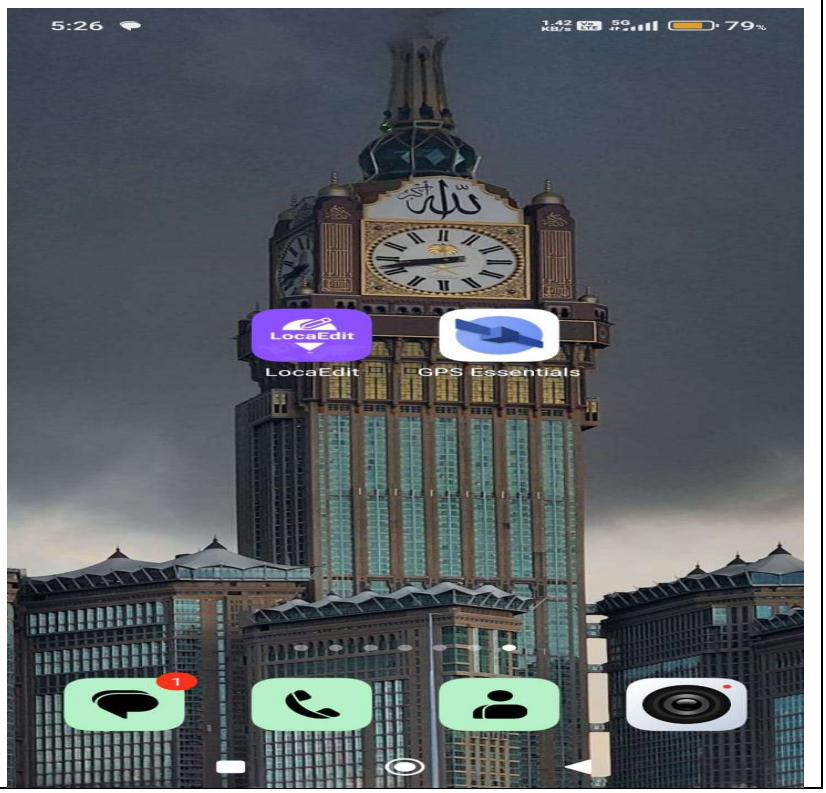
Session	**2025-26 (ODD)**	**Course Name**	**PE-I - Geo-Intelligence for Smart IoT Devices Lab**
Semester	**5**	**Course Code**	**23IOT1523**
Roll No	**64**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**06**
Course Outcome	Apply and demonstrate the use of proprietary and open-source GIS tools (e.g., QGIS) for creating, visualizing, and managing spatial datasets.
Aim	Collect real time GPS coordinates (5–10 points) using a mobile GPS app and import into QGIS.
Problem Definition	Every student has to use their own ideology here.
Theory (100 words)	LocaEdit is a network configuration management tool used in telecom systems to maintain accurate cell information such as frequencies, antenna details, and neighbor relations. It helps engineers edit and update site data to improve coverage quality and handover performance. On the other hand, GPS Essentials is a mobile application used for field surveys and drive testing. It records location data like coordinates, speed, and route tracks to link network performance with geographic positions. While LocaEdit works at the planning level to ensure correct network parameters, GPS Essentials supports real-time data collection in the field. Together, both tools contribute to efficient network optimization and monitoring.

Procedure and Execution (100 Words)	Implementation Steps: 1. Download GpsEssential from playstore 2. Turn your Gps on and move to location you want to track. 3. Open Gps essential : a. Go to STREAMS b. Click on All Tracks. c. Click the plus '+' icon. d. A new window will appear named Create Stream. e. Navigate to Tracks. f. Click on START. 4. Export the Track in KML format. 5. Once all Tracks are export, Open Qgis application.Import the OpenStreetMap.Now import a layer and select the Kml file we exported. Once the file is imported you can see the Tracks we recorded on Qgis. 

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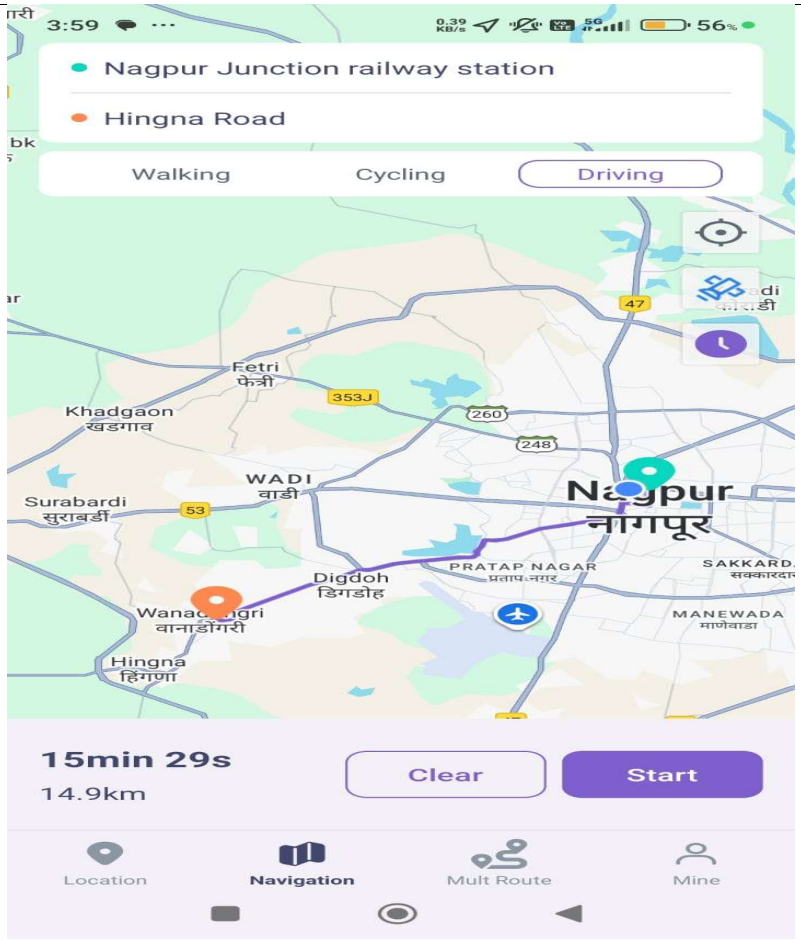

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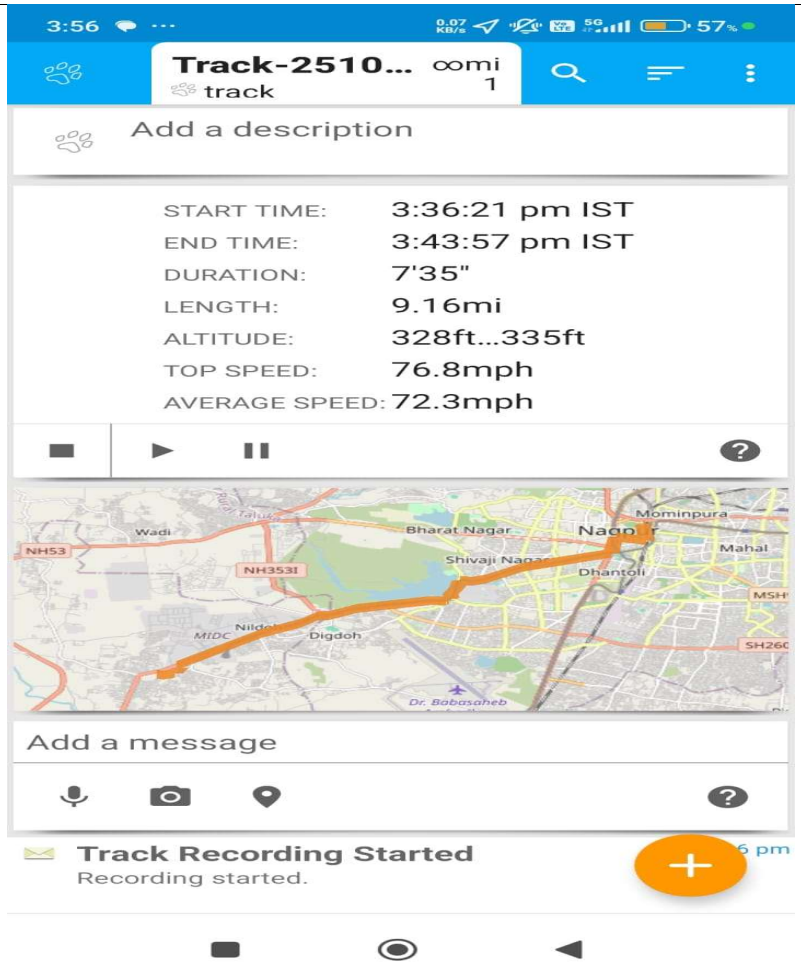

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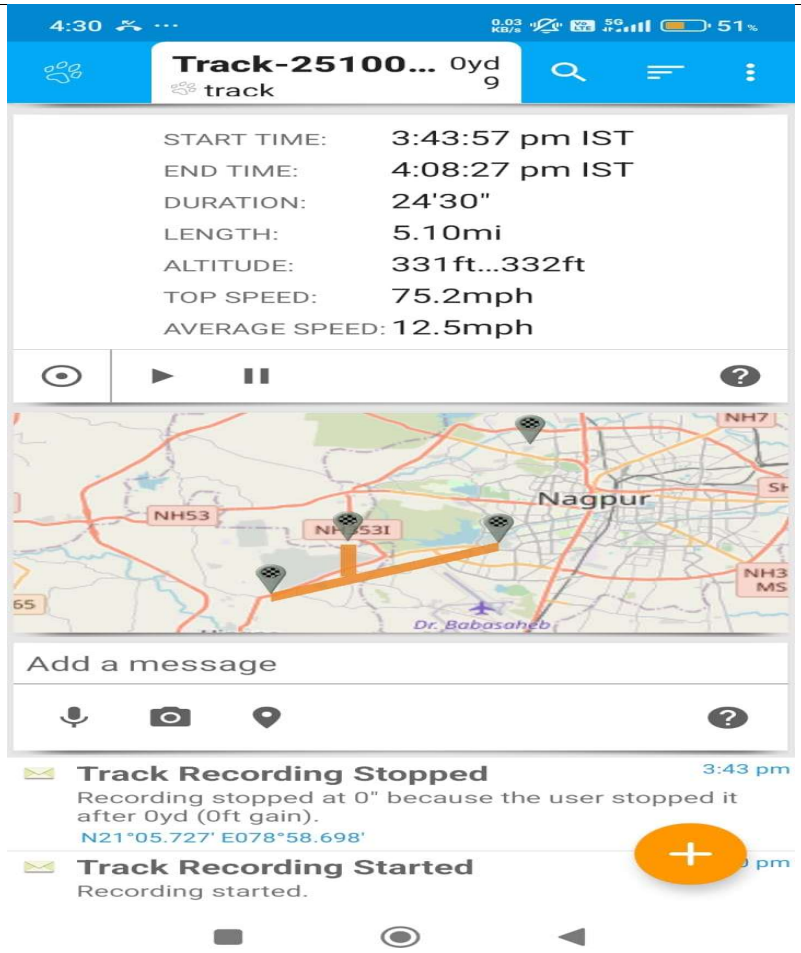

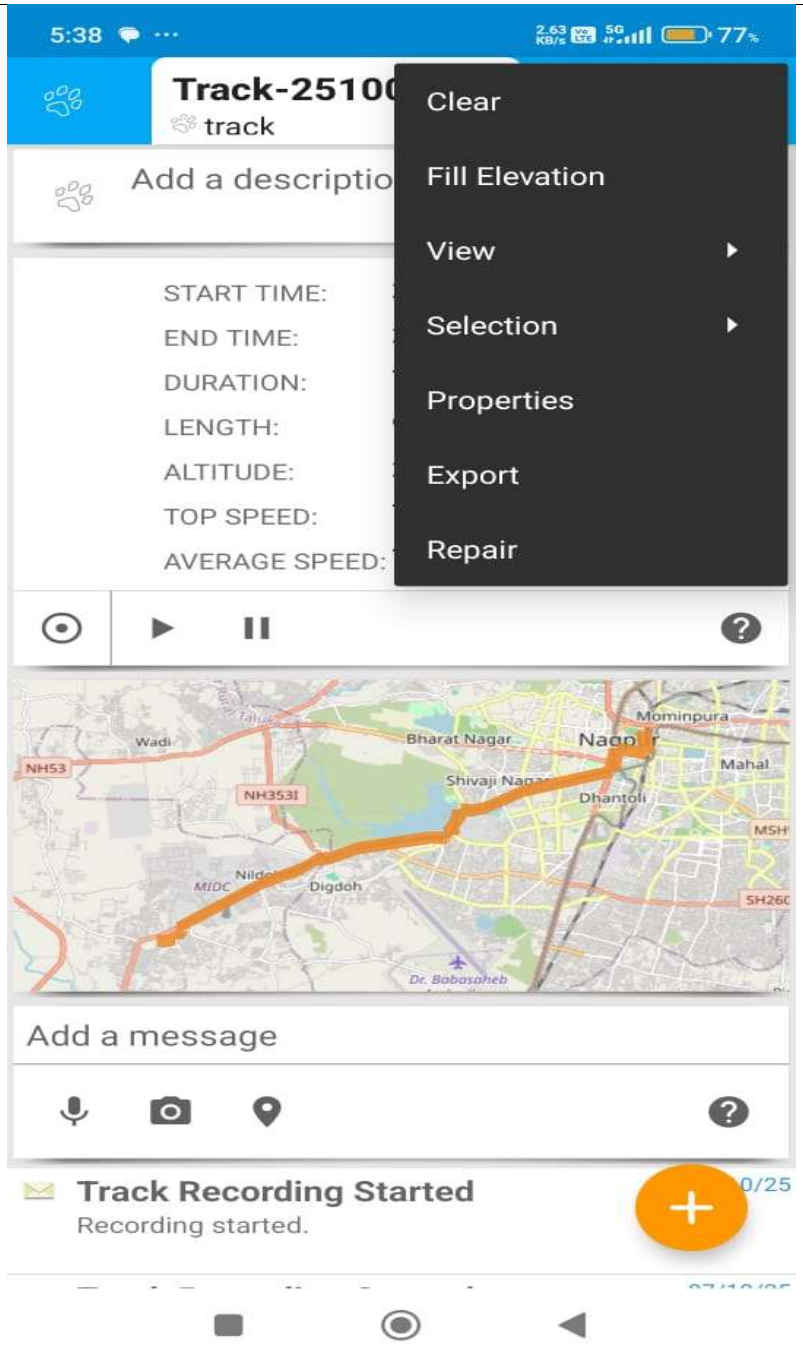
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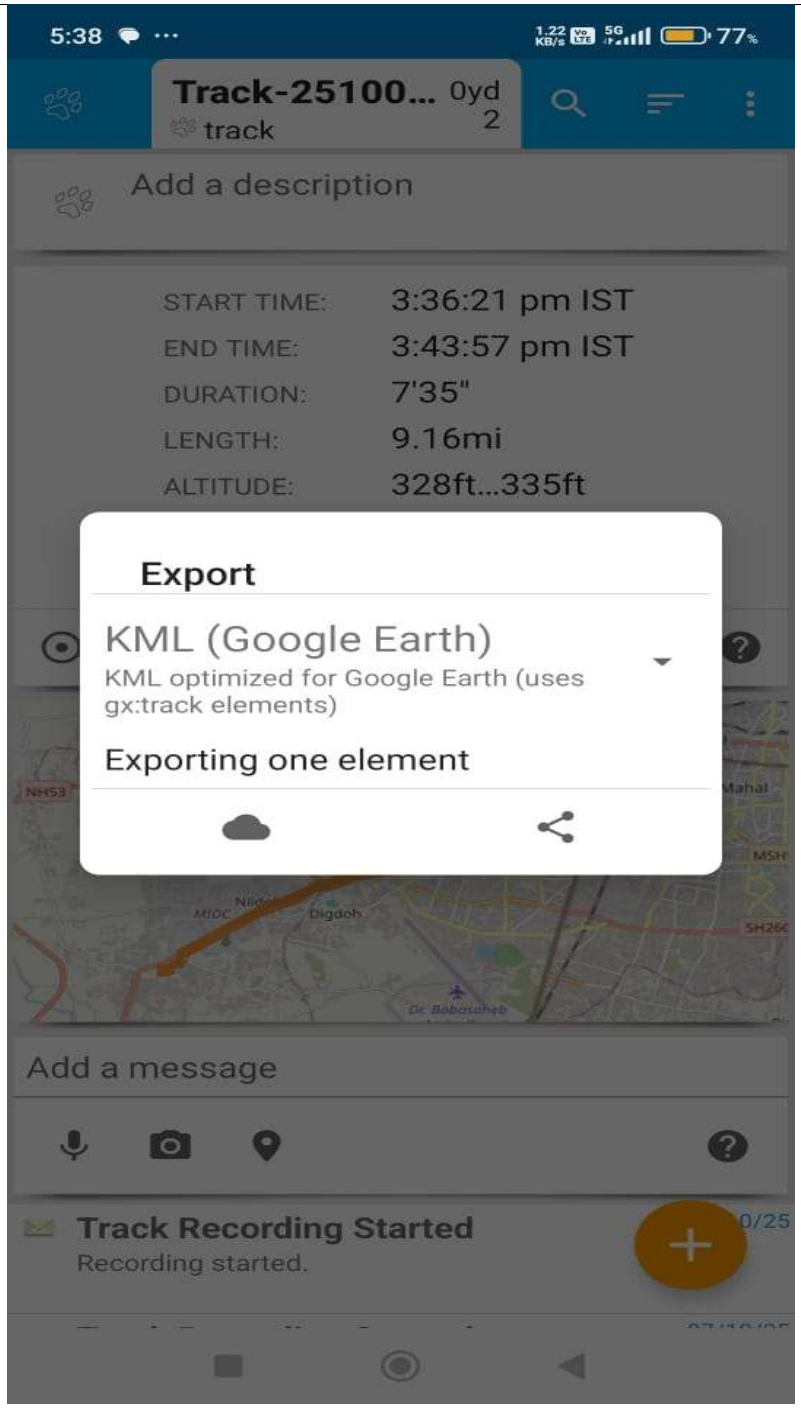
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KML (Google Earth)
KML optimized for Google Earth (uses gx:track elements)
Exporting one element
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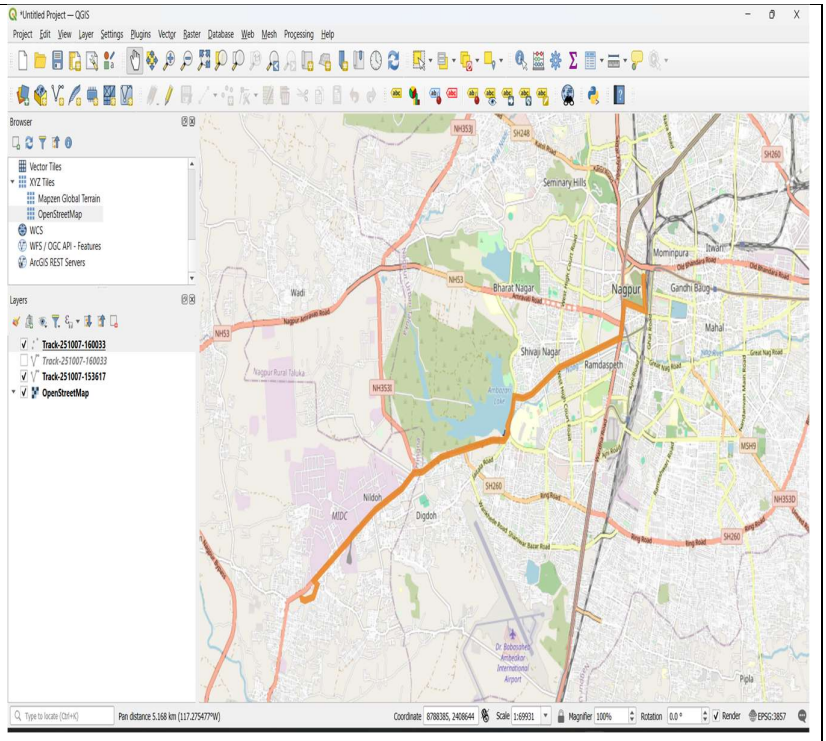

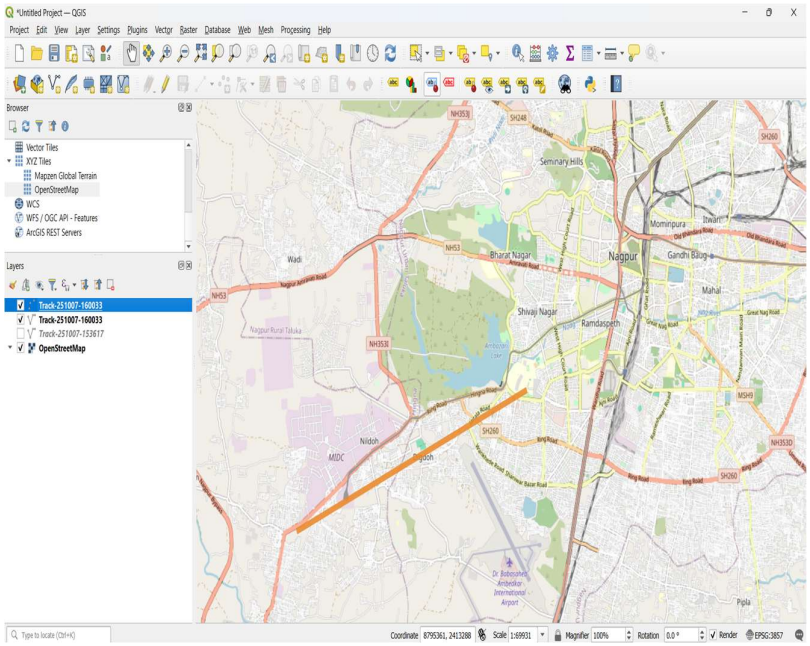

Output Analysis	The GPS locations of Nagpur Railway Station and YCCE College were plotted and visualized in QGIS to study their spatial context within the city. Nagpur Railway Station appeared in the highly developed central zone, surrounded by dense commercial activities and major transport routes, indicating strong accessibility and high foot traffic. In contrast, YCCE College is located in the outer part of Nagpur near Hingna Road, where development density is comparatively lower. By visualizing these points on the map, it became easier to understand the difference in connectivity and urban influence between the two locations. This analysis helps evaluate accessibility, planning needs, and the distribution pattern of important landmarks across Nagpur.
Conclusion	The practical of collecting real-time GPS coordinates using a mobile GPS app and importing them into QGIS successfully demonstrated the process of acquiring, managing, and visualizing spatial data. By plotting the collected points on the map, it became easier to understand their exact geographical positions and spatial distribution in the real world. This exercise improved practical skills in field data collection, coordinate handling, and GIS mapping techniques. Overall, the practical highlighted the importance of GPS and QGIS in accurate location analysis, urban planning, navigation studies, and decision-making based on geospatial information.
Plag Report (Similarity index < 12%)	
Date	**07/10/2025**